

Culvert Screening Rules and Source Hierarchy

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1. Purpose and scope

This procedure defines the source hierarchy, calculation rules, validation checks, priority screening logic, and traceability requirements for reconciling stormwater culvert asset records. It applies only to the three input files named in Table 1. The procedure is a screening workflow for asset management and does not replace a detailed hydraulic design study, structural inspection, or regulatory flood assessment.

Table 1. Input files controlled by this procedure

File	Primary role	Key fields used
stormwater_culvert_asset_register.csv	Base asset register and inspection record	asset_id; culvert_shape; culvert_material; diameter_mm; span_mm; rise_mm; barrel_count; culvert_length_m; upstream_invert_m; downstream_invert_m; register_slope_percent; observed_blockage_percent; structural_condition_score; hydraulic_condition_score; reported_overtopping; recent_backwater_depth_m; maintenance_status; last_cleaned_date
storm_event_and_maintenance_log.csv	Storm response, work-order, maintenance closure, CCTV, and geometry follow-up evidence	log_id; asset_id; log_date; record_type; storm_event_id; road_overtopping_confirmed; observed_backwater_depth_m; observed_blockage_percent; work_order_status; action_completed_date; post_action_blockage_percent; cctv_required; cctv_completed_date; source_reliability_rank
culvert_field_observation_index.csv	Independent field observation and dimension evidence	observation_id; asset_id; observation_date; measured_diameter_mm; measured_span_mm; measured_rise_mm; measurement_quality;

File	Primary role	Key fields used
		barrel_accessibility; inlet_blockage_percent; outlet_blockage_percent; field_high_water_mark; field_overtopping_evidence; field_backwater_depth_m; immediate_safety_concern

2. Source hierarchy and tie-break rules

Apply the highest ranked applicable source in Table 2. If a higher ranked source is not applicable to the value being reconciled, use the next ranked source. Do not use external data, assumed design standards, or professional judgement to fill missing values unless this procedure provides a rule.

Table 2. Source hierarchy

Rank	Source and applicable fields	Use rule
1	culvert_field_observation_index.csv for measured dimensions, field blockage, high-water mark, overtopping evidence, backwater depth, safety concern, and visible damage	Use when observation_date is available. For dimensions, use only when measurement_quality is Good or Fair and the measured dimension field is populated. Use the latest observation for the same metric unless a later closed maintenance record updates current blockage.
2	storm_event_and_maintenance_log.csv for storm-response overtopping, work-order status, post-action blockage, CCTV follow-up, and maintenance closure	Use for storm-event confirmation and maintenance chronology. A Closed work order with action_completed_date and post_action_blockage_percent updates current blockage after that date unless a later valid field observation is available.
3	stormwater_culvert_asset_register.csv for base asset identity, geometry, register slope, inspection values, and maintenance status	Use as the base source when Rank 1 or Rank 2 evidence does not correct, derive, or supersede a value.

Table 3. *source_reliability_rank* used inside *storm_event_and_maintenance_log.csv*

Rank value	Evidence type	Interpretation
1	Engineering technician or CCTV contractor evidence	Highest reliability within <i>storm_event_and_maintenance_log.csv</i> . Use for same-date conflicts when the same metric is reported by multiple log records.
2	Storm patrol or field crew storm-response evidence	Second reliability level. Use for storm-event overtopping and observed backwater unless a Rank 1 record for the same metric and date exists.
3	Maintenance crew, maintenance inspector, scheduler, or route inspection record	Third reliability level. Use for maintenance closure and work-order status. Closed work orders still control current blockage when the chronology rule applies.
4	Screened public complaint	Lowest reliability level. Use as follow-up evidence only unless confirmed by Rank 1, Rank 2, or applicable field-observation evidence.

Tie-break rules

- **TH-01:** For repeated records from the same source file and the same metric, use the latest dated record unless TH-02, TH-03, or TH-06 applies.
- **TH-02:** A confirmed storm overtopping value remains part of the storm-risk history even when a later maintenance closure reduces blockage.
- **TH-03:** For current blockage, a later Closed work order with *action_completed_date* and *post_action_blockage_percent* supersedes earlier inspection or storm blockage values.
- **TH-04:** A Scheduled or Open work order does not change current blockage unless *post_action_blockage_percent* is present.
- **TH-05:** Blank cells, Unknown values, and access-limited field observations do not justify assumed values. Classify the affected item as unresolved follow-up when the required value cannot be derived.
- **TH-06:** When *storm_event_and_maintenance_log.csv* contains same-date records for the same asset and same metric, use the lower *source_reliability_rank* value. If dates differ, apply TH-01 through TH-04 before using *source_reliability_rank*.

3. Geometry and slope calculation rules

Use SI units in all calculations. Convert millimetres to metres before area calculations. Report calculated values to two decimal places unless more precision is needed for audit.

Table 4. Geometry and slope rules

Rule ID	Culvert shape or metric	Calculation or decision rule
GE-01	Circular	$\text{effective_open_area_m2} = \text{barrel_count} \times 3.1416 \times (\text{diameter_mm} / 1000)^2 / 4$. A circular culvert requires <code>diameter_mm</code> from the asset register or <code>measured_diameter_mm</code> from the field-observation file.
GE-02	Box	$\text{effective_open_area_m2} = \text{barrel_count} \times (\text{span_mm} / 1000) \times (\text{rise_mm} / 1000)$. A box culvert requires <code>span_mm</code> and <code>rise_mm</code> from the asset register or measured field dimensions.
GE-03	Pipe arch	$\text{effective_open_area_m2} = \text{barrel_count} \times \text{arch_area_screening_factor} \times (\text{span_mm} / 1000) \times (\text{rise_mm} / 1000)$, where <code>arch_area_screening_factor</code> = 0.82 for this screening procedure only.
GE-04	Field measurement use	A <code>measured_diameter_mm</code> , <code>measured_span_mm</code> , or <code>measured_rise_mm</code> in <code>culvert_field_observation_index.csv</code> may derive a missing asset-register dimension only when <code>measurement_quality</code> is Good or Fair and the measured value is present. <code>barrel_accessibility</code> = Limited does not prevent use of a populated measured value, but it cannot justify assuming a blank value.
GE-05	Unresolved geometry	If a required dimension remains blank after applying GE-01 through GE-04, classify the asset as unresolved geometry follow-up and do not calculate <code>effective_open_area_m2</code> .
SL-01	Slope	$\text{computed_slope_percent} = ((\text{upstream_invert_m} - \text{downstream_invert_m}) / \text{culvert_length_m}) \times 100$, using values from <code>stormwater_culvert_asset_register.csv</code> .
SL-02	Slope validation	Flag slope as inconsistent when <code>computed_slope_percent</code> is less than 0 or when $\text{abs}(\text{computed_slope_percent} - \text{register_slope_percent})$ is greater than 0.25 percentage points.

4. Blockage, backwater, and overtopping rules

Table 5. Blockage, backwater, and overtopping rules

Rule ID	Topic	Decision rule
BL-01	Current blockage source	Use the latest valid source for current_blockage_percent in this order: (1) latest valid field observation from culvert_field_observation_index.csv; (2) if later than the field observation, post_action_blockage_percent from storm_event_and_maintenance_log.csv when work_order_status is Closed and action_completed_date is present; (3) if neither source is available, observed_blockage_percent from stormwater_culvert_asset_register.csv. For field observations, use the larger of inlet_blockage_percent and outlet_blockage_percent.
BL-02	Blockage class	Classify current_blockage_percent as Low when <20, Moderate when 20 to 39, High when 40 to 59, and Severe when >=60.
BW-01	Backwater source	Use field_backwater_depth_m from culvert_field_observation_index.csv when present. Otherwise use observed_backwater_depth_m from storm_event_and_maintenance_log.csv. Otherwise use recent_backwater_depth_m from stormwater_culvert_asset_register.csv.
BW-02	Backwater class	Classify backwater as Low when <0.40 m, Moderate when 0.40 to 0.74 m, High when 0.75 to 0.99 m, and Critical when >=1.00 m.
OT-01	Overtopping source	Classify reconciled_overtopping_status as Confirmed when road_overtopping_confirmed is Yes in storm_event_and_maintenance_log.csv or field_overtopping_evidence is Yes in culvert_field_observation_index.csv.
OT-02	Overtopping correction	When reported_overtopping in stormwater_culvert_asset_register.csv is No or Unknown and OT-01 gives Confirmed, correct the reconciled overtopping value to Yes and trace the source log_id or observation_id.
OT-03	Unresolved overtopping	When backwater_class is High or Critical and reconciled_overtopping_status is No or Unknown with no direct confirmation from road_overtopping_confirmed or field_overtopping_evidence, flag the asset for overtopping follow-up rather than changing overtopping to Yes.

5. Condition, work-order, and priority classification rules

Table 6. Condition, work-order, and priority classification rules

Rule ID	Topic	Decision rule
CN-01	Condition score	Interpret structural_condition_score and hydraulic_condition_score from stormwater_culvert_asset_register.csv on a 1 to 5 scale where 1 is very poor and 5 is good.
CN-02	Poor condition	Classify structural condition as Poor when structural_condition_score <=2. Classify hydraulic condition as Poor when hydraulic_condition_score <=2.
WO-01	Closed work order	A Closed work order in storm_event_and_maintenance_log.csv is accepted as completed only when action_completed_date is present.
WO-02	Open or scheduled work order	Open or Scheduled work_order_status is unresolved follow-up when current_blockage_percent >=40, overtopping is Confirmed, cctv_required is Yes, or immediate_safety_concern is Yes.
PR-01	Priority 1 - Emergency	Assign Priority 1 when immediate_safety_concern is Yes, or when overtopping is Confirmed and backwater is Critical, or when overtopping is Confirmed and current_blockage_percent >=60.
PR-02	Priority 2 - High	Assign Priority 2 when Priority 1 is not met and any condition is true: overtopping is Confirmed; backwater is High; current_blockage_percent >=60; hydraulic condition is Poor with current_blockage_percent >=40; or cctv_required is Yes with Open work_order_status.
PR-03	Priority 3 - Medium	Assign Priority 3 when Priority 1 and Priority 2 are not met and any condition is true: current_blockage_percent is 40 to 59; backwater is Moderate; unresolved geometry follow-up exists; or structural_condition_score is 3 with outlet_erosion_class Moderate or Severe.
PR-04	Priority 4 - Low	Assign Priority 4 when no Priority 1, Priority 2, or Priority 3 condition applies.
PR-05	Priority tie-break	When more than one priority rule applies, use the highest urgency class: Priority 1 before Priority 2 before Priority 3 before Priority 4.

6. Validation exception rules

Table 7. Validation exception triggers

Rule ID	Exception name	Flag condition
VR-01	Missing geometry	Required diameter_mm, span_mm, or rise_mm remains blank after applying GE-01 through GE-04.
VR-02	Slope conflict	SL-02 identifies negative computed_slope_percent or a slope difference greater than 0.25 percentage points.
VR-03	Overtopping correction	OT-02 changes reported_overtopping from No or Unknown to Yes using log or field evidence.
VR-04	Overtopping follow-up	OT-03 applies because backwater is High or Critical without direct overtopping confirmation.
VR-05	High blockage	current_blockage_percent >=60, or current_blockage_percent >=40 with backwater High or Critical.
VR-06	Open work order risk	WO-02 applies for an Open or Scheduled work order.
VR-07	Safety concern	immediate_safety_concern is Yes in culvert_field_observation_index.csv.
VR-08	Asset-register status conflict	maintenance_status is No action or Monitor in stormwater_culvert_asset_register.csv while Priority 1 or Priority 2 is assigned from higher ranked evidence.

Table 8. Validation exception resolution status

Rule ID	Resolution status
VR-01	Unresolved follow-up unless a later survey record resolves the missing required dimension.
VR-02	Unresolved follow-up unless a later survey record resolves the slope conflict.
VR-03	Resolved correction.
VR-04	Unresolved follow-up.
VR-05	Unresolved follow-up unless a later Closed work order reduces current_blockage_percent below 40.
VR-06	Unresolved follow-up.
VR-07	Unresolved follow-up.
VR-08	Unresolved follow-up.

7. Reconciled output requirements

The analysis output must keep corrected, derived, accepted, and unresolved values separate. Do not overwrite a source value without retaining the source value and reconciliation status.

Table 9. Required workbook sheets

Workbook sheet	Required content
Reconciled Asset Register	One row per asset_id from stormwater_culvert_asset_register.csv, with source values, reconciled values, status fields, and traceability references.
Geometry Calculations	Area and slope calculations; source dimensions; computed_slope_percent; slope validation result; formula or text calculation expression.
Source Conflict Log	Rows where Rank 1, Rank 2, or Rank 3 evidence differ for overtopping, blockage, dimensions, maintenance status, or backwater.
Validation Exceptions	All VR-01 through VR-08 exceptions triggered by the three input files, with asset_id, rule_id, source file, source record ID, value, and status.
Priority Classification	Priority 1 through Priority 4 classification, rule_id responsible for classification, and supporting source record ID.
Maintenance Worklist	Unresolved follow-up items, open or scheduled work orders, emergency/high-priority assets, and required action category.
Executive Summary	Record count, derived geometry count, overtopping corrections, unresolved follow-up count, priority-class counts, and highest-priority assets.

8. Traceability requirements

Table 10. Traceability requirements

Traceability item	Requirement
Accepted source value	Report source_file, source_record_id, source_field, accepted_value, and status = Accepted.
Corrected value	Report source_file, source_record_id, source_field, corrected_value, rule_id, and status = Corrected.
Derived value	Report calculation_expression, input fields, source_file for each input field, source_record_id, derived_value, rule_id, and status = Derived.
Validation flag	Report asset_id, rule_id, triggering value, threshold, source_file, source_record_id, and status = Resolved or Unresolved follow-up.
Priority class	Report asset_id, priority_class, priority_rule_id, supporting source_file, supporting source_record_id, and concise reason.
No external source	Do not use maps, rainfall atlases, municipal databases, design manuals, or assumed site measurements outside the three input files named in Table 1.

9. Memo requirements

The memo must summarize the method, key corrected or derived values, highest priority assets, unresolved follow-up items, and traceability basis. It must not introduce values from outside the three input files. Numerical values should be rounded consistently and should not display raw floating-point artifacts.

10. Document status

This procedure defines deterministic screening rules for the associated asset-register, event-log, and field-observation files. It is intended for an audit-style data reconciliation exercise using only the named input files.